Data-Driven ADHD Diagnosis: A Comparative Analysis Using PSD and SVM

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***Abstract*—Attention-Deficit Hyperactivity Disorder (ADHD) is a neurodevelopmental condition that impacts an individual's ability to focus, regulate impulses, and manage activity levels. It typically manifests in childhood but can persist into adulthood, often affecting daily activities such as school and work. Diagnosing ADHD is particularly challenging due to the subtle differences in brain function and behavior, which frequently overlap with symptoms of other conditions. This study utilizes a dataset comprising three groups—healthy controls, ADHD-inattentive, and ADHD-hyperactive patients—with 190 features derived from various brain regions. The data was preprocessed and analyzed using Power Spectral Density (PSD) to extract meaningful features, capturing frequency-domain information essential for distinguishing between the groups. The processed features were then used as input for a Support Vector Machine (SVM), a machine learning model well-suited for classification tasks. All data preprocessing and feature extraction were conducted using MATLAB. The focus of this work is a comparative analysis to assess the contribution of the dataset's attributes to prediction accuracy and to investigate how the model's accuracy varies across different feature ranges. The findings highlight that while the model demonstrates robust performance for smaller feature ranges, its accuracy diminishes as the number of features increases. This underscores the importance of careful feature selection and optimization in machine learning applications for ADHD classification.**

***Index Terms*—ADHD, PSD, SVM, Dataset**

I. Introduction

Attention-deficit hyperactivity disorder (ADHD) is a prevalent neurodevelopmental condition marked by symptoms such as excessive inattention, hyperactivity, and impulsivity that are not age-appropriate [1]. While the symptoms of ADHD in children tend to diminish with age, they often persist into adulthood, with two-thirds of affected children continuing to experience impairments if not adequately treated [2]. Early and proper diagnosis of ADHD is crucial [3]. Currently, there is no definitive biological test for diagnosing ADHD; instead, it is identified through psychiatric assessments of behavioral symptoms [3]. Functional magnetic resonance imaging (fMRI), known for its non-invasive nature and high resolution, can identify brain abnormalities in individuals with ADHD, offering objective insights to aid clinical psychiatric diagnosis [4]. fMRI captures changes in deoxyhemoglobin concentration within the brain's local blood vessels, both during task performance and in resting states [6]. Quantitative features derived from fMRI, either voxel-based or region-of-interest (ROI)-based, provide insights into local brain activity and the connectivity between brain regions [7]. Metrics such as regional homogeneity (ReHo), amplitude of low-frequency fluctuations (ALFF), and fractional ALFF (fALFF) are calculated at the voxel level [7]. ReHo quantifies the similarity or synchronization of time series within a spatial cluster [8]. In ROI-wise analysis, the entire brain is initially divided into multiple regions of interest (ROIs) based on an anatomical or functional brain atlas [9]. Numerous diagnostic models for ADHD based on fMRI imaging have been developed utilizing machine learning techniques [5].

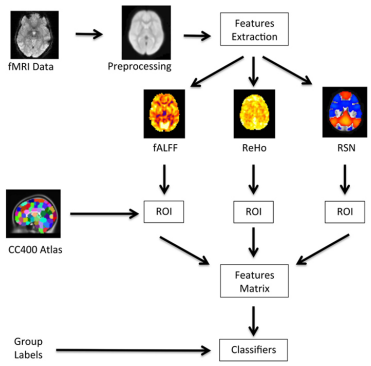


Fig. 1. Data processing pipeline: The pre-processing of the raw fMRI data [10].

Correlation-based analyses can uncover variations in functional connectivity patterns across different groups [14]. Since low-frequency fluctuations are crucial for identifying correlations between brain regions and resting-state networks, their magnitude can differ across individuals and brain regions [11]. The Fractional Amplitude of Low-Frequency Fluctuations (fALFF) measures the amplitude of low-frequency oscillations. It is calculated as the ratio of the amplitude within the low-frequency range to the total amplitude across the entire observable frequency range of a given signal [11]. Again, Power Spectral Density (PSD) analysis aids in identifying and measuring low-frequency fluctuations in the BOLD signal, which are commonly linked to resting-state networks and spontaneous brain activity [12]. Features derived from PSD can be utilized in machine learning models to distinguish brain states or differentiate between clinical population and healthy

individuals [13].

As there is currently no definitive clinical diagnostic process for ADHD, this research aims to predict the likelihood of belonging to one of the affected groups, offering a more robust diagnostic approach compared to traditional behavioral evaluations. This gap in diagnosis motivated me to conduct this research, focusing on identifying significant differences in data that could serve as a standard for diagnosing the disorder. The approach involves collecting a large dataset and using machine learning techniques to analyze the intricate data structure and predict outcomes. By utilizing this methodology, the study seeks to contribute to the development of a reliable and data-driven diagnostic work for ADHD.

II. Research Design and Methods

The dataset utilized in this research comprises preprocessed fMRI data from three distinct groups: 748 healthy controls, 260 ADHD-inattentive patients, and 175 ADHD-hyperactive patients, with information derived from 190 defined brain regions. A comprehensive preprocessing pipeline was applied, incorporating steps such as head motion correction, spatial smoothing, normalization, frame-wise displacement adjustment, and temporal band-pass filtering [15]. These preprocessing techniques effectively minimized artifacts and ensured standardization across subjects and groups, facilitating reliable comparisons.

The 190 brain regions were identified through spectral clustering, a data-driven method applied to resting-state fMRI data. This advanced approach enabled the segmentation of the brain into functionally coherent and anatomically meaningful regions, providing a robust framework for further analysis. By integrating thorough preprocessing and precise regional definition, the dataset establishes a strong foundation for exploring functional connectivity patterns and group-specific differences in brain activity.

A. Feature Extraction

The data analysis and feature extraction were performed using the Power Spectral Density (PSD) technique, implemented in MATLAB. This approach facilitates the extraction of frequency-domain features, ensuring robust and reliable analysis of the fMRI data. Data processing to prepare the dataset for machine learning, Power Spectral Density (PSD) technique was employed. PSD analysis enables a comprehensive examination of the signal dynamics within specific brain regions, providing valuable insights into the correlations and functional connectivity between different brain areas [16]. This method ensures that the extracted features capture the essential frequency-domain characteristics of the brain's activity, enhancing the dataset's suitability for machine learning applications.

To develop a standardized dataset with features derived from Power Spectral Density (PSD), a structured methodology was implemented. This process encompassed key steps, including data preprocessing, feature extraction, augmentation, normalization, and integration. The dataset was composed of three distinct groups: healthy individuals, ADHD-inattentive patients, and ADHD-hyperactive patients, with the data retrieved from the data.mat file.

The initial step focused on processing the raw data for each patient in the dataset. Each patient's data was assumed to be a time-series representation of multiple features, organized in a cell array format. To compute the PSD for each feature, Welch's method was utilized. This technique is a widely recognized approach for estimating the power spectral density of a signal , expressed as [17]:

$$\mathrm{S}_x(f) = \frac{1}{N} \sum_{i=1}^{N} |X_i(f)|^2$$

Here, Xi(f) denotes the Fourier Transform of the i-th segment of the signal, while N represents the total number of overlapping segments. This approach minimizes noise in the PSD estimation by averaging across the segments, resulting in a smoother representation in the frequency domain.

The mean of the computed PSD was used as a representative value for each feature of the patient's data. This method consolidates the spectral information into a single metric, effectively capturing the power distribution in the frequency domain. The mean PSD values for all features were arranged into a matrix, with rows representing individual patients and columns corresponding to specific features. This process yielded a processed dataset for each group, ensuring consistency and uniform representation across patients.

B. Augmentation to Achieve Uniform Feature Dimensions

To ensure compatibility with machine learning algorithms and maintain consistency across datasets, the number of features for each patient was expanded to a target dimension of 950. This feature augmentation process involved appending statistical descriptors derived from the original feature set. Specifically, variance (Var(x)), standard deviation (Std(x)), skewness (Skew(x)), and kurtosis (Kurt(x)) were computed for each patient's features. These descriptors provide higher-order statistical insights into the data distribution, thereby enriching the representation of each patient's feature set.

C. Data Normalization

Normalization was applied to each dataset to scale all feature values to a consistent and comparable range. This step utilized min-max normalization, ensuring that the relative relationships between features were preserved while standardizing their values. The equation [18]:

$$\mathrm{x}_{\mathrm{norm}} = \frac{x - \min(x)}{\max(x) - \min(x) + \epsilon}$$

where epsilon is a small constant to prevent division by zero. Normalization ensures that all features contribute equally to subsequent analyses, preventing features with large ranges from dominating the results.

D. Padding for Standardization

To account for differences in the number of patients across datasets, padding was applied by adding rows of zeros to smaller datasets, aligning their patient count with the largest dataset. This approach ensures seamless integration of datasets into a unified structure for subsequent processing and analysis without introducing bias.

E. Dataset Integration

The datasets representing three classes—healthy individuals, ADHD-inattentive patients, and ADHD-hyperactive patients—underwent processing that included feature augmentation, normalization, and padding to ensure uniformity. The processed data was organized into separate matrices and saved as a MATLAB file for subsequent use in machine learning and statistical analyses. The final dataset preserves the frequency-domain characteristics extracted through Welch's method, ensuring both consistency and scalability across all groups, thereby making it suitable for further analytical applications.

III. Results

The systematic application of Power Spectral Density (PSD) analysis and feature augmentation ensures the dataset captures the critical spectral characteristics of the original signals while maintaining compatibility with computational analysis workflows. Following that the dataset was employed as input to a machine learning model, specifically a Support Vector Machine (SVM). This model classifies the dataset by determining the optimal hyperplane that separates the classes (e.g., healthy, ADHD-inattentive, ADHD-hyperactive) based on the extracted features [19]. The hit rate is computed as the percentage of accurately classified instances, providing a measure of the model's performance.

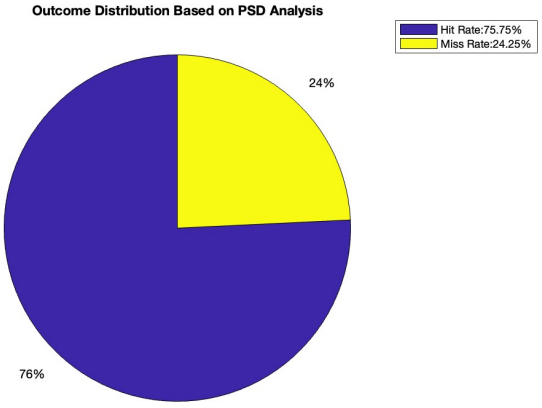


Fig. 2. Model prediction outcome distribution measuring PSD with 190 features

The results indicate varying levels of accuracy depending on the number of features used in the model.

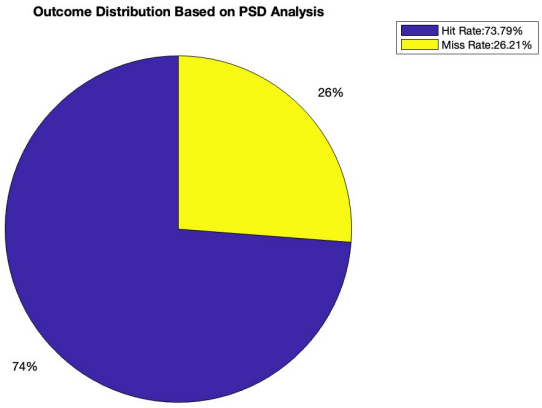


Fig. 3. Model prediction outcome distribution measuring PSD with 380 features

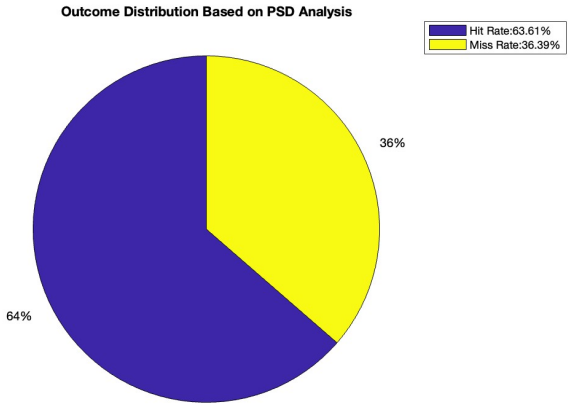


Fig. 4. Model prediction outcome distribution measuring PSD using 950 features

IV. Analysis and Interpretation

The results indicate that an increase in the number of features negatively impacts the model's accuracy. This decline may be attributed to the model overlooking critical information or incorporating irrelevant features, which can dilute the overall performance and hinder accurate predictions. The processes of augmenting the dataset and subsequently normalizing it have the potential to impact the originality of the data. These steps, while essential for preparing the dataset for machine learning, may inadvertently alter its inherent characteristics, potentially influencing the outcomes of the analysis. Diagnosing ADHD presents significant challenges due to the subtle differences observed within brain regions among the patients. These minimal variations make it difficult to distinguish between the groups, further complicating the

diagnostic process [20].

The current method is designed to work specifically with datasets containing 950 features. It cannot directly handle datasets with more features due to the fixed augmentation strategy. To overcome this limitation, the program can be redesigned to dynamically adapt to datasets with varying feature dimensions, making it more versatile and applicable to a broader range of scenarios.

V. Conclusion

In conclusion, this research highlights an important observation from the machine learning model that utilized the Power Spectral Density (PSD) technique. The model demonstrated strong performance when operating within a specific range of features but showed a decline in accuracy when processing a larger number of features. This suggests that the model may overlook critical information or give undue attention to less relevant details, ultimately impacting its predictive ability. The key finding of this research is that applying Power Spectral Density (PSD) analysis enables the model to achieve better prediction accuracy when operating on a smaller range of features.

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